CLINICAL · EAR

Vestibular testing: bedside and laboratory

Vestibular tests tell you which side is weak and whether the problem is peripheral or central; they never tell the whole story. Normal results do not exclude vestibular disease, so every test should be chosen and read with the history in mind. The bedside examination comes first, and often it is enough.

AT A GLANCE

- Bedside tests come first: nystagmus, head impulse, Dix-Hallpike, gait.
- Calorics test the lateral canal; VEMP tests the otoliths.
- Fixation failure and direction change suggest central.
- Normal tests don't exclude vestibular disease.
- vHIT gain ≥0.8 is normal; catch-up saccades mean a weak canal; oVEMP is the best screen for dehiscence.

1 CHAPTER ONE Overview

What the balance system is made of.

1.1 The end organs

The three semicircular canals sense angular (rotational) acceleration; the two otolith organs, utricle and saccule, sense linear acceleration and head tilt.

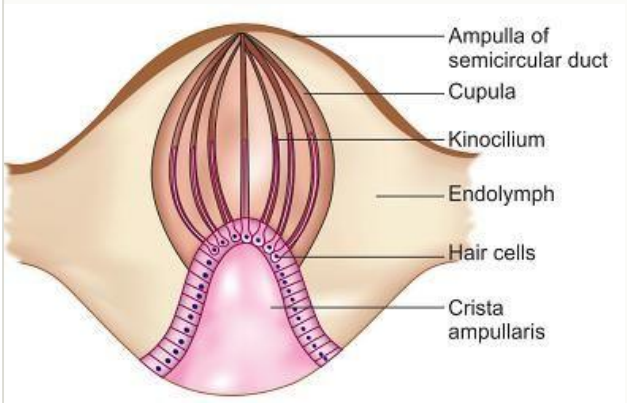


Fig. 41: Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

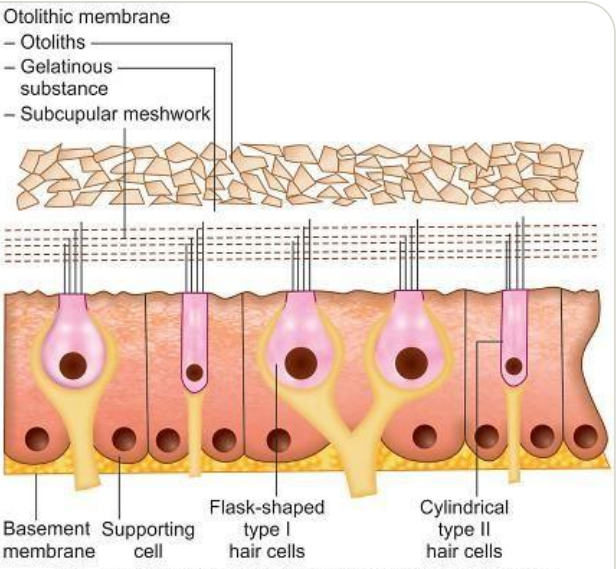


Fig. 43: Macula of otolith organs: utricle and saccule

FIGURE 1 The crista ampullaris of a canal, and the macula of an otolith organ.

Anatomy atlas · ENT Steps p. 78, ENT Steps p. 80

1.2 The nerves

The *superior vestibular nerve* supplies the lateral and superior canals and the utricle; the *inferior vestibular nerve* supplies the posterior canal and the saccule.

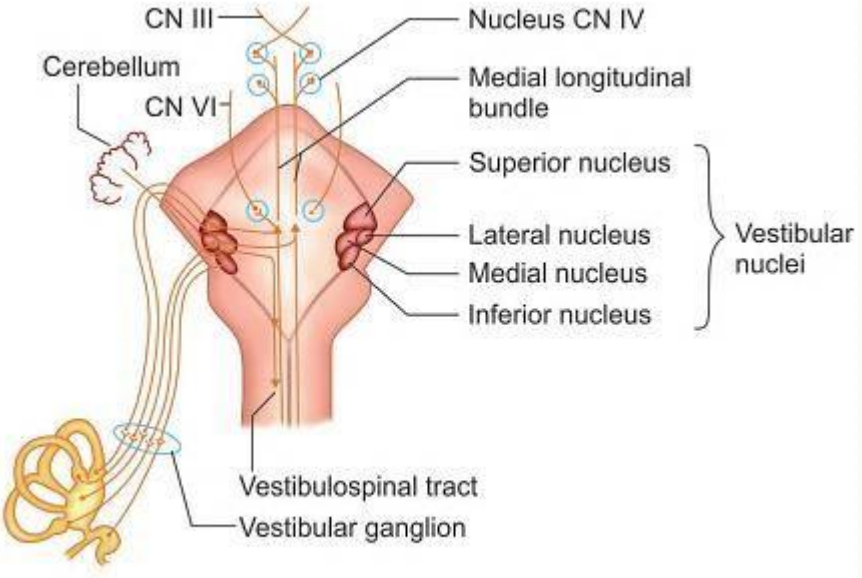


Fig. 49: Vestibular pathways

FIGURE 2 Central vestibular pathways.

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1.3 Physiology to remember

At rest both labyrinths fire equally. If the right labyrinth fails, the brain reads the imbalance as turning to the left: the eyes drift slowly to the right and flick back to the left. So nystagmus beats *away* from a dead labyrinth (a paretic lesion) and *towards* an irritated one.

1.4 Ewald's laws

The eyes move in the plane of the stimulated canal. In the lateral canal, flow towards the ampulla (ampullopetal) stimulates more than flow away from it; in the vertical canals the reverse is true.

1.5 What each test measures

WHAT EACH TEST MEASURES

TEST	MEASURES
Caloric test	Each lateral canal, at low frequency
Rotary chair	Both lateral canals together
Video head impulse test	Each of the six canals, at high frequency
Cervical VEMP	Saccule and inferior vestibular nerve
Ocular VEMP	Utricle and superior vestibular nerve
Posturography	How the patient uses vestibular, visual and proprioceptive input

2 CHAPTER TWO Bedside examination

First, and often enough.

2.1 Spontaneous nystagmus

Look in the primary position and in each direction of gaze, with and without fixation (Frenzel glasses remove fixation). Peripheral nystagmus is horizontal-torsional, suppressed by fixation, and stronger looking towards the fast phase.

2.1.1 Degrees (Alexander's law)

- **First degree:** only when looking towards the fast phase.
- **Second degree:** also looking straight ahead.
- **Third degree:** in all directions of gaze.

2.1.2 Central patterns

- **Downbeat:** craniocervical junction (Chiari malformation, cerebellar degeneration).
- **Upbeat:** brainstem.
- **Pure torsional:** brainstem.
- **Pendular:** multiple sclerosis and congenital nystagmus.

2.2 Head impulse test

A rapid head turn; a catch-up saccade shows a failed vestibulo-ocular reflex on that side.

2.3 Dynamic visual acuity

Losing more than five lines on a chart while the head is shaken suggests bilateral vestibular loss.

2.4 Positional tests

2.4.1 Dix-Hallpike manoeuvre

With the patient sitting, turn the head 45 degrees to one side, then lie them back quickly with the head 30 degrees below horizontal. Watch for latency, duration, direction and fatigue.

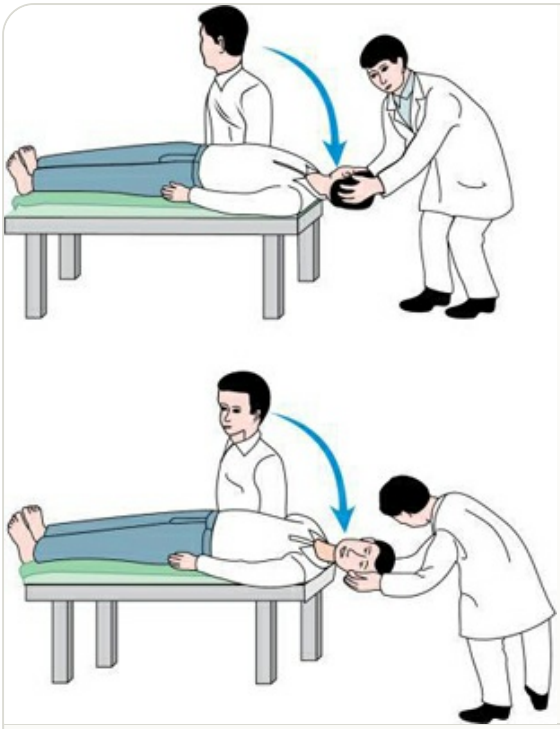


FIGURE 3 The Dix-Hallpike manoeuvre.
Dhingra, fig. 6.1

POSITIONAL NYSTAGMUS

FEATURE	PERIPHERAL	CENTRAL
Latency	2–20 seconds	None
Duration	Under a minute	Over a minute
Direction	Fixed, towards the undermost ear	Changes
Fatigue	Yes	No
Vertigo	Severe	None or slight

2.5 Fistula test

Pressure on the tragus or with a Siegle speculum causes nystagmus when the labyrinth is open: lateral canal erosion, a fenestration cavity, superior canal dehiscence, or a window leak. It needs a working labyrinth: a dead ear, or

cholesteatoma sealing the fistula, gives a false negative. *Hennebert's sign* is a positive test without a fistula, in congenital syphilis and some Ménière's disease.

2.6 Balance and cerebellar tests

Romberg, tandem walking, Unterberger's stepping test, and past-pointing: falling and past-pointing go towards the side of the slow phase. Cerebellar hemisphere disease causes dysmetria, dysdiadochokinesia and rebound; midline disease causes a wide gait and truncal ataxia.

3 CHAPTER THREE Laboratory tests

Measuring each part.

3.1 Caloric test

Warm or cold water (or air) in the canal sets up convection currents in the lateral canal. With the patient lying and the head raised 30 degrees, the lateral canal is vertical and responds best.

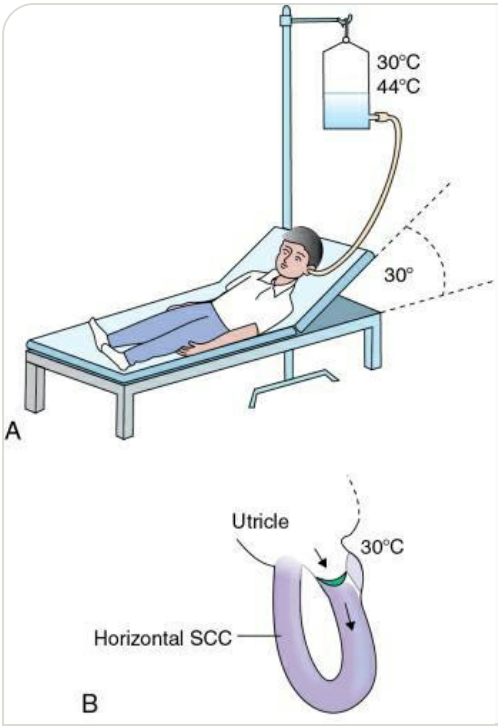


FIGURE 4 The Fitzgerald-Hallpike caloric test: the head is raised 30 degrees to make the lateral canal vertical.

Dhingra, fig. 6.2

3.1.1 COWS

Cold water makes nystagmus beat to the opposite side; warm to the same side.

3.1.2 Reading it

- **Canal paresis:** one side responds less (a difference of more than about 15 to 30%).
- **Directional preponderance:** nystagmus stronger in one direction (more than about 25 to 30%).
- **Bilateral weakness:** bilateral vestibular loss.

3.1.3 A worked example

Each ear is irrigated with warm (44 °C) and cool (30 °C) water or air, and the peak slow-phase velocity of the nystagmus is measured in degrees per second. **Jongkees' formula** compares the two sides: canal paresis = [(RW + RC) − (LW + LC)] ÷ (RW + RC + LW + LC) × 100. For example, with right warm 30, right cool 28, left warm 12 and left cool 10°/s, the result is (58 − 22) ÷ 80 × 100 = 45%, a left canal paresis (most labs use a cut-off of about 20 to 25%). Directional preponderance compares right-beating (RW + LC) with left-beating (LW + RC) responses in the same way. If the sum of all four responses is very small (under about 6 to 8°/s per irrigation), the result is bilateral weakness, confirmed by rotary chair or head

impulse testing. Failure of fixation to suppress the nystagmus (fixation index) points to a central lesion.

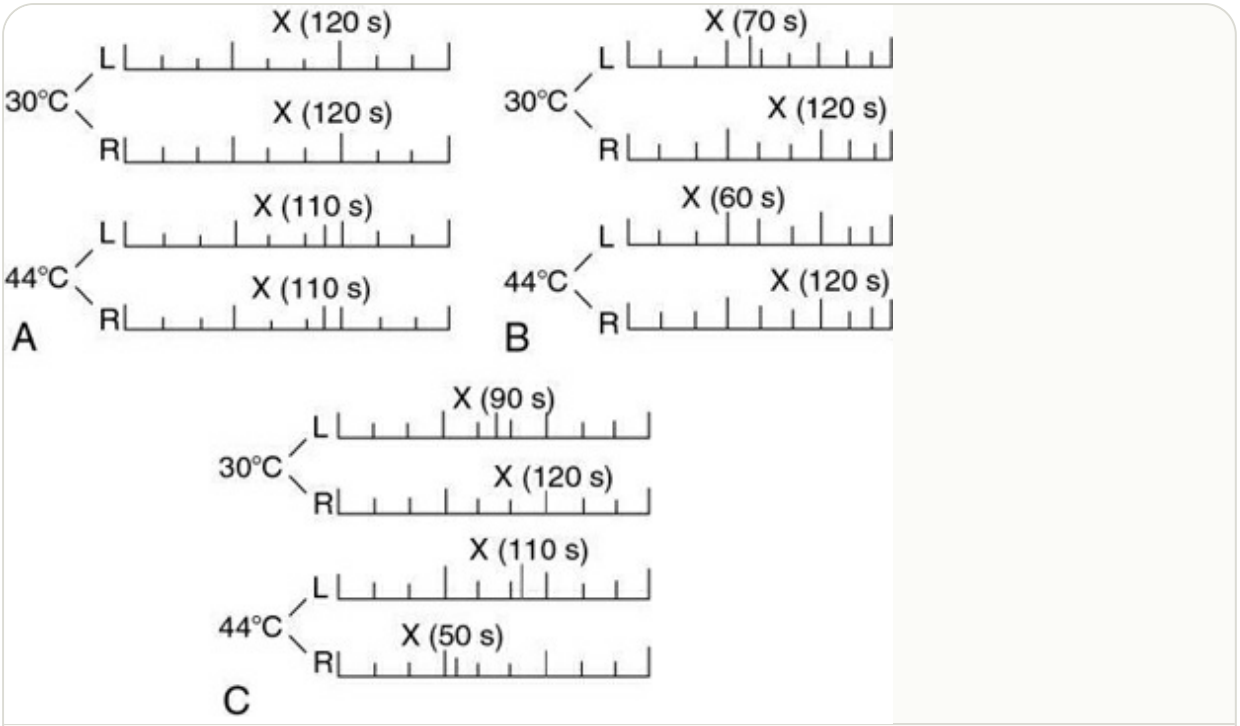


FIGURE 5 Calorigrams: (A) normal, (B) left canal paresis, (C) left directional preponderance. Dhingra, fig. 6.3

WHERE THE BOOKS DIFFER · THE CALORIC TEST

Dhingra	Lee
Water at 30°C and 44°C for 40 seconds.	For 30 seconds; considers directional preponderance of questionable value.

3.1.4 Kobrak test

Small volumes of iced water, increasing from 5 mL; no response suggests a dead labyrinth.

CALORICS AND PERFORATIONS

Don't use water with a perforation; use air.

3.2 Electronystagmography and videonystagmography

Record eye movements during calorics, positional tests, saccades, pursuit and optokinetic testing. Central signs: direction-changing nystagmus, failure of fixation

suppression, abnormal saccades and pursuit, and hyperactive calorics.

3.3 Video head impulse test

Measures the gain of the reflex for each canal separately; low gain with overt or covert saccades means a weak canal.

Small, fast, unpredictable head turns are made while a lightweight camera records the eye. The **gain** is eye velocity divided by head velocity: normal for the lateral canals is about 0.8 or more (vertical canals about 0.7). A weak canal has low gain and **catch-up saccades**: *overt* saccades after the head stops, which the examiner can see at the bedside, and *covert* saccades during the movement, which only video can detect. The test is quick, tests all six canals, and measures the high frequencies of head movement used in daily life, whereas calorics test only the lateral canal at a very low frequency; the two are complementary, and Ménière's disease often shows abnormal calorics with a normal vHIT. A normal vHIT in acute continuous vertigo is a warning sign of a central cause.

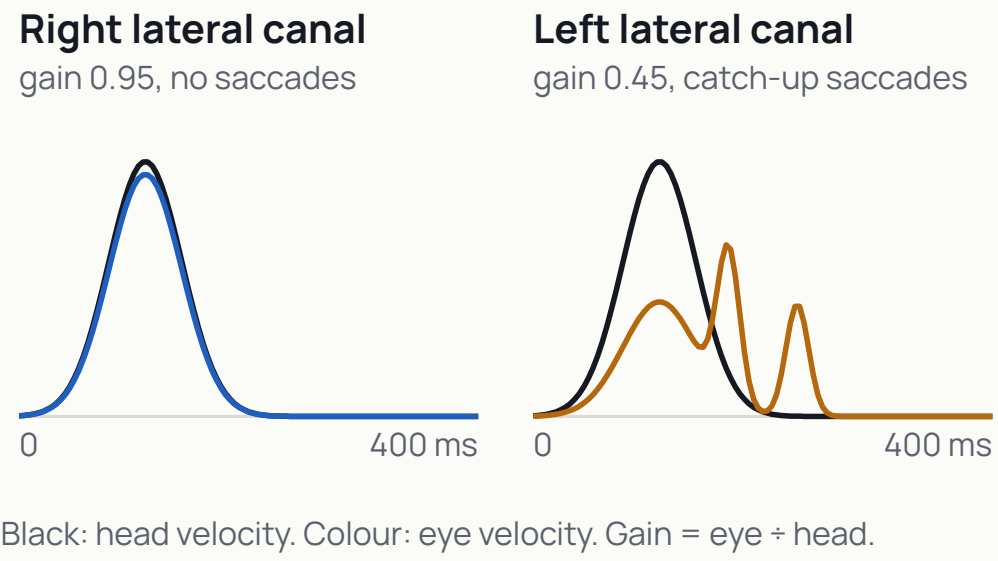


FIGURE 6 Reading a video head impulse test. On the right the eye matches the head (gain near 1). On the left the eye lags (gain 0.45) and catch-up saccades follow: a left lateral canal weakness, as in vestibular neuritis.

Drawn for ENT Atlas

3.4 Vestibular evoked myogenic potentials

Loud sounds stimulate the saccule (cervical VEMP from sternocleidomastoid) and utricle (ocular VEMP from the inferior oblique). They work even in deaf ears. In superior canal dehiscence thresholds are abnormally low and ocular amplitudes high.

READING VEMPS

FINDING	CERVICAL VEMP (SACCULE, INFERIOR VESTIBULAR NERVE)	OCULAR VEMP (UTRICLE, SUPERIOR VESTIBULAR NERVE)
Superior canal dehiscence	Low threshold (often under 70 dB nHL)	Large amplitude, low threshold: the best screening test
Vestibular neuritis, superior nerve	Usually preserved	Absent on the affected side
Vestibular neuritis, inferior nerve	Absent on the affected side	Preserved
Ménière's disease	Reduced or absent, or tuning shifted to higher frequencies	Variable
Vestibular schwannoma	Often reduced or absent	Often reduced or absent
Conductive loss	Air-conducted VEMP lost; bone-conducted VEMP preserved	Same

3.5 Rotary chair

Tests both labyrinths together over a range of frequencies; useful in bilateral loss, in children, and when a canal cannot be irrigated.

3.6 Posturography

Shows how the patient uses each sense to balance; useful in rehabilitation and in non-organic imbalance.

4 CHAPTER FOUR Using the tests

When to order them.

- When the side or site of the lesion is unclear.
- Before a destructive procedure, to check the other ear.
- Planning rehabilitation and tracking recovery.
- Monitoring gentamicin effect; daily head impulse and visual acuity help detect toxicity.

PEARLS

- A canal paresis with directional preponderance away from it suggests Ménière's disease; preponderance towards the same side, a schwannoma.
- Fixation suppressing the nystagmus is peripheral; failure to suppress suggests a central cause.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 On call

THE DIZZY PATIENT AT THE BEDSIDE

1. Spontaneous nystagmus, with and without fixation.
2. Head impulse test and test of skew.
3. Gait and cerebellar signs.
4. Dix-Hallpike when the neck and condition allow.

5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Ewald's laws** and why nystagmus beats away from a dead ear.
- **Dix-Hallpike** technique and peripheral versus central positional nystagmus.
- **Caloric test:** technique, COWS, canal paresis and directional preponderance.
- **Fistula test** and Hennebert's sign.
- **VEMP** and what it tests.

5.3 Common pitfalls

COMMON PITFALLS

- Using water calorics with a perforation.
- Taking normal tests as proof of no vestibular disease.
- Missing central signs because the eyes were tested only with fixation.

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